Universe creation by tunnelling from laboratory data: no manifold instanton, a unique canonical exponent, and a Smarr identity

Project Baby Universe — Genesis Branch

21 October 2026 (Stage One manuscript, draft); addenda 22–23 October 2026

Abstract

We revisit the thin-wall model of a false-vacuum bubble (de Sitter interior, Schwarzschild exterior) and the proposal of Farhi, Guth and Guven that such a bubble, too small to inflate classically, may tunnel into an inflating, causally disconnected daughter universe. (i) We prove that, for every transition starting from regular, time-symmetric, asymptotically flat ("laboratory") data, the Euclidean exterior interpolating region cannot be embedded in Euclidean Schwarzschild. The pseudomanifold encountered by Farhi–Guth–Guven is therefore forced. The obstruction is tied to a sign change of the exterior extrinsic curvature while the wall recedes from the Euclidean horizon; an interior sign change does not obstruct embedding. (ii) In the reduced canonical theory with the asymptotic clock, the branch of the Euclidean momentum is fixed by analyticity. The competing branch is a horizon monodromy, which in Euclidean language is the half-turn slicing jump. (iii) At $M = 0$ the exponent is $B = S_E[\text{instanton}] + \frac{1}{4G}(\sum A_{\max} - \sum A_{\min})$ in closed form; it equals the Coleman–De Luccia exponent of the false vacuum when the false vacuum is the larger cap. (iv) For $M > 0$ the exponent obeys a first law $d\tilde{B} = (\mathrm{Vol}_E^{\mathrm{in}}/8\pi G)\, d\Lambda_{\mathrm{in}} + A_E^W\, d\sigma - \Delta t_E^W\, dM$ and a Smarr identity $B = -\frac{\Lambda_{\mathrm{in}}}{8\pi G}\mathrm{Vol}_E^{\mathrm{in}} - \frac{\sigma}{2}A_E^W - \frac{M}{2}\Delta t_E^W + \frac{\mathcal{A}}{4G}$. This is a covariant expression in which the pseudomanifold enters only through one boundary term. Numerically, over laboratory transitions $B \geq 0.56\, B(M \to 0)$ for all parameters and $B \geq B_{\mathrm{CdL}}$ when $\chi \gtrsim 1.2\kappa$; for GUT-scale parameters and gram-scale bubbles $B \sim 10^6$. All statements are verified symbolically and numerically with an open-source code.

1 Introduction

A spherical region of false vacuum separated from true vacuum by a domain wall can have an inflating interior even though, seen from outside, it is a black hole [1]. Farhi and Guth showed that such a configuration cannot arise classically from regular initial data when the null energy condition holds [2]. Farhi, Guth and Guven (FGG) asked whether quantum tunnelling could do it [3]. Their canonical calculation was hampered by the choice of slicing, and their Euclidean interpolation was not a manifold but a "pseudomanifold" with a conjectured action. Fischler, Morgan and Polchinski found agreement within a Hamiltonian treatment [4]. Later Hamiltonian work extended these transitions to all signs of the cosmological constant [7, 8]. Holographic arguments suggest that smooth saddles do not produce inflating bubbles [9, 10].

This paper makes the FGG difficulties precise and fixes the branch of the exponent within the reduced canonical theory (see the Addendum: a residual ambiguity survives in the single-flip class). Section 2 fixes the model and conventions. Section 3 summarises the global structure. Section 4 proves the embedding obstruction. Section 5 fixes the branch of the canonical exponent. Sections 6–7 give the closed form at $M = 0$, the first law and the Smarr identity. Section 8 collects numerical results and Section 9 the limitations.

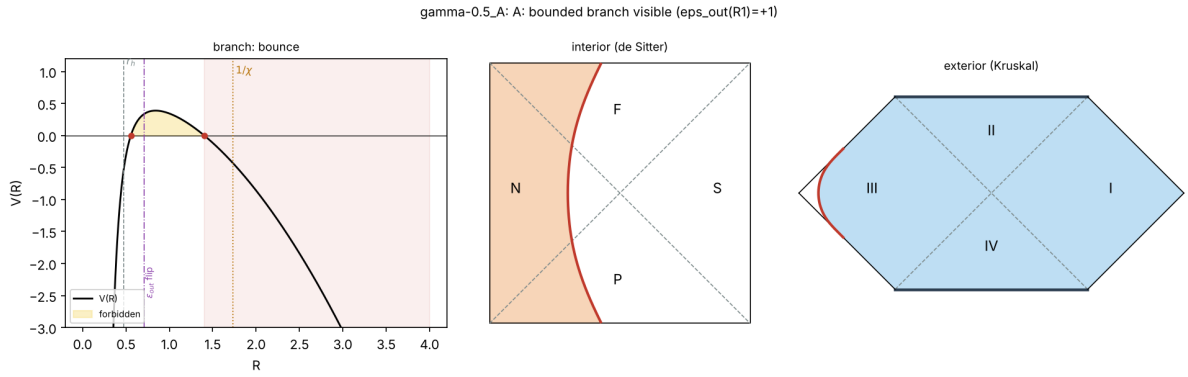


Figure 1: Inflating bounce branch ($\gamma = -0.5$, regime A): effective potential, interior de Sitter diagram, exterior Kruskal diagram, all drawn from integrated trajectories.

2 Model and conventions

Signature $(-,+,+,+)$, $c = 1$. Interior $f_{\rm in} = 1 - \chi^2 r^2$, exterior $f_{\rm out} = 1 - 2GM/r - \eta^2 r^2$ (here mostly $\eta = 0$), wall $S_{ab} = -\sigma h_{ab}$, $\kappa = 4\pi G\sigma$. With n pointing from interior to exterior, $K_{ab} = e^\mu_a e^\nu_b \nabla_\mu n_\nu$, $[X] = X_+ - X_-$, the Israel conditions $[K_{ab}] - h_{ab}[K] = -8\pi G S_{ab}$ reduce to

$$A_{\rm in} - A_{\rm out} = \kappa R, \qquad A \equiv n^r = \varepsilon\sqrt{f + \dot R^2}, \tag{1}$$

with $K_{\theta\theta} = RA$ and $K^\tau{}_\tau = (\ddot R + \frac{1}{2}f')/A$. Using the identity $A_{\rm in}^2 - A_{\rm out}^2 = f_{\rm in} - f_{\rm out}$,

$$A_{\rm in} = \frac{f_{\rm in} - f_{\rm out} + \kappa^2 R^2}{2\kappa R}, \quad A_{\rm out} = \frac{f_{\rm in} - f_{\rm out} - \kappa^2 R^2}{2\kappa R}, \quad \dot R^2 + V(R) = 0, \quad V = f_{\rm out} - A_{\rm out}^2. \tag{2}$$

Lemma 1. *For $\sigma > 0$, $(R, \varepsilon_{\rm in}, \varepsilon_{\rm out})$ solves the unsquared junction condition iff $\dot R^2 + V = 0$ and $\varepsilon = \mathrm{sgn} A(R)$ on each side. The only spurious solutions of the squared equation are those with $\sigma < 0$.*

With $R = (2GM/L^2)^{1/3} z$, $L^4 = (\chi^2 - \eta^2 + \kappa^2)^2 + 4\kappa^2\eta^2$: $V = 1 - g(z)/E$, $g = z^2 - 2\gamma/z + z^{-4}$, $\gamma = (\chi^2 - \eta^2 - \kappa^2)/L^2$, $E = 4\kappa^2 L^{-8/3}(2GM)^{-2/3}$, and for $\eta = 0$, $\varepsilon_{\rm out} = \mathrm{sgn}(1 - z^3)$, $\varepsilon_{\rm in} = \mathrm{sgn}(1 - \gamma z^3)$.

3 Global structure

For $M < M_{\rm cr} = 4\kappa^3/(G g_{\rm min}^{3/2} L^4)$ there are a bounded branch $(0 \to R_1 \to 0)$ and a bounce branch $(\infty \to R_2 \to \infty)$.

Proposition 2. *(i) $\varepsilon_{\rm out}(R_2) = -1$ always. (ii) $\varepsilon_{\rm out}(R_1) = +1$ iff $E > 2 - 2\gamma$ (regime A). (iii) At $E = 2 - 2\gamma$, $R_1 = 2GM$. (iv) In regime A, the flip radius R_f with $A_{\rm out}(R_f) = 0$ satisfies $R_1 < R_f < R_2$.*

The causal region occupied by the wall is determined by $(\mathrm{sgn} f, \varepsilon, \mathrm{sgn}\dot R)$. We implemented this coordinate-free itinerary and verified it against explicit integration in Kruskal and global de Sitter coordinates: all 15 branches of our atlas agree (Fig. 1). Every inflating branch ends in region III, is invisible from region I, and has a past-singular exterior, in accordance with [2]. The time-symmetric slice at R_2 contains the Einstein–Rosen bridge, a minimal sphere. Hence every classical history through it is singular to the past and to the future, and a transition to it must replace its past.

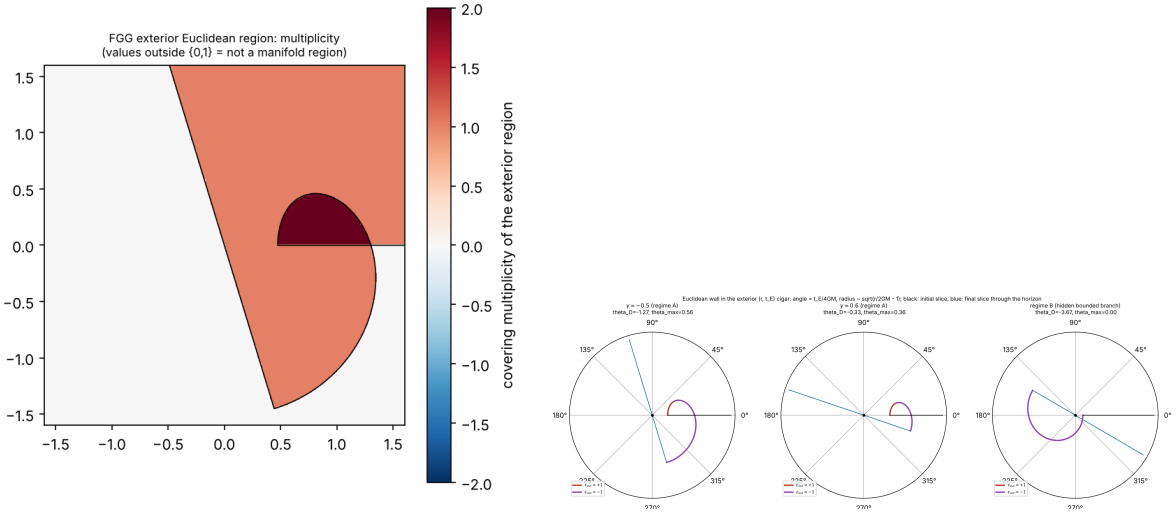


Figure 2: Left: covering multiplicity of the exterior Euclidean region for a regime-A transition. Right: the Euclidean wall in the cigar for three models.

4 The exterior embedding obstruction

Let $\mathcal{C}$ be the (r, t_E) section of Euclidean Schwarzschild (a smooth cigar), with polar coordinates (ρ, θ), $\theta = t_E/4GM$. The Euclidean wall obeys $R'^2 = V$, $f_{\rm out}\, dt_E/d\tau_E = A_{\rm out}(R)$.

Theorem 3 (no manifold instanton)**.** *Suppose that (H1) the half-bounce runs from R_1 to $R_2 > R_1$ with $R' > 0$; (H2) $A_{\rm out}$ has a single simple zero in (R_1, R_2) with $A_{\rm out}(R_1) > 0 > A_{\rm out}(R_2)$; (H3) the initial and final exterior slices are the totally geodesic slices ($\theta = 0$ ray; broken line through the tip at angles $\theta_D, \theta_D + \pi$); (H4) the exterior region is an oriented surface locally isometric to $\mathcal{C}$, with boundary initial slice $\cup$ wall $\cup$ final slice. Then the local isometry is not injective.*

Proof sketch. Orientation places the region outside the $\varepsilon = +1$ arc, inside the $\varepsilon = -1$ arc, counterclockwise of the initial ray, and clockwise of the final ray. If the map were injective, the multiplicity would be 1 on the region, 0 off it, and would jump by one across each boundary piece. Monotonicity of R puts the $\varepsilon = -1$ arc outside the $\varepsilon = +1$ arc. If either arc wraps by $\geq 2\pi$, a radial segment between consecutive windings has multiplicities 0 and 1 at its ends with no crossing between them. Otherwise, at infinity the multiplicity is 1 exactly on the counterclockwise arc $(0, a)$, $a \equiv \theta_D + \pi$, while just outside the $\varepsilon = -1$ arc it is 0. A case analysis in θ_D produces in each case an angle at which these two requirements conflict. The full proof is in the companion file `NO_GO_THEOREMS.md`. □

For the dS|Schwarzschild model, every regime-A transition satisfies (H1)–(H4). Degree-theory multiplicity maps confirm the result: they take values outside $\{0, 1\}$ (Fig. 2), including negative values for small M, where the wall winds dozens of times around the horizon. Regime-B transitions ($\varepsilon_{\rm out} = -1$ throughout) are also non-manifold numerically, with $|\theta_D| > \pi$ over the whole regime. The interior is not obstructed: models with an $\varepsilon_{\rm in}$ flip have embedded interiors, because there the wall approaches the Euclidean horizon.

5 The canonical exponent and its branch

With the asymptotic Killing time as clock and $H = M$, Hamilton's equation $\partial P/\partial M|_R = dt/dR$ gives

$$P = \frac{R}{G}\Big[\operatorname{artanh}\frac{\dot R}{A_{\rm out}} - \operatorname{artanh}\frac{\dot R}{A_{\rm in}}\Big], \qquad 2\sqrt{f_{\rm in}f_{\rm out}}\cosh\frac{GP}{R} = \varepsilon_{\rm in}\varepsilon_{\rm out}(f_{\rm in}+f_{\rm out}-\kappa^2R^2), \quad (3)$$

together with $\partial P/\partial\Lambda_{\rm in}|_{R,M} = -\frac{1}{8\pi G}\frac{4\pi R^3}{3}\frac{dt_{\rm in}}{dR}$ and $\partial P/\partial\sigma|_{R,M} = -4\pi R^2\frac{d\tau}{dR}$. Under the barrier $P = iP_E$, $P_E = \frac{R}{G}(\Theta_{\rm out}-\Theta_{\rm in})$ with $(A,\sqrt{V}) = \sqrt{f}(\cos\Theta,\sin\Theta)$, and $B = 2\int_{R_1}^{R_2} P_E\,dR$.

Proposition 4 (branch selection)**.** *On (R_1,R_2) we have $f \geq V > 0$ on both sides, so P is analytic there. Its branch points are the turning points and the horizons $f = 0$; the flip radius is a regular point. The WKB phase is the analytic continuation along the barrier, $\Theta = \operatorname{atan2}(\sqrt{V},A) \in [0,\pi]$. The principal branch differs by a horizon monodromy ($P \to P \pm i\pi R/G$, $t \to t \pm i\beta/2$), which is the Euclidean half-turn slicing jump of FGG. The principal branch gives $B < 0$ in 41 of 45 test models.*

6 Closed form at $M = 0$

Theorem 5. *Let $M = 0$, $R_0 = 2\kappa/L^2$, $p = -A_{\rm out}(R_0)$, $q = A_{\rm in}(R_0)$ ($p+q = \kappa R_0 > 0$). Then*

$$S_E = -\frac{\pi R_0^2}{2G}\Big[\frac{1}{1+p}+\frac{1}{1+q}\Big], \qquad B = \frac{\pi R_0^2}{2G}\frac{p+q+2p_+q_+}{(1+|p|)(1+|q|)} = S_E + \frac{1}{4G}\Big(\sum A_{\max} - \sum A_{\min}\Big), \quad (4)$$

the sums running over the extremal spheres of the instanton's time-symmetric slice. If $q < 0$, then $B = B_{\rm CdL}$, the decay exponent of the false vacuum.

The proof uses $J(s) = \int_0^1 x\operatorname{atan2}(\sqrt{1-x^2},sx)dx = \pi/4(1+s)$ ($s \geq 0$), $\pi/2 - \pi/4(1-s)$ ($s < 0$), together with the cap volumes. Gravitational $O(4)$ bounces of a quartic scalar converge to $B_{\rm CdL}$ as $O(w)$ in the wall width (0.18% at $w = 0.005$).

7 First law and Smarr identity

Theorem 6. *Let $\tilde B = B - \mathcal{A}/4G$, where $\mathcal{A}$ collects the endpoint areas ($\pm 4\pi R_i^2$ where the wall is an extremal sphere of its slice, plus $4\pi/\chi^2$ if the final slice contains the de Sitter equator). Then, within a regime,*

$$d\tilde B = \frac{\mathrm{Vol}_E^{\rm in}}{8\pi G}d\Lambda_{\rm in} + A_E^W\,d\sigma - \Delta t_E^W\,dM, \qquad B = -\frac{\Lambda_{\rm in}}{8\pi G}\mathrm{Vol}_E^{\rm in} - \frac{\sigma}{2}A_E^W - \frac{M}{2}\Delta t_E^W + \frac{\mathcal{A}}{4G}, \quad (5)$$

where $\mathrm{Vol}_E^{\rm in}$ is the Euclidean interior 4-volume of the full bounce, A_E^W the wall's Euclidean area, and Δt_E^W the exterior Euclidean Killing time along the wall.

Proof. The first law is the Euclidean continuation of the three Hamilton–Jacobi relations, with endpoint contributions $2P_E(R_i)dR_i = d(\mathcal{A}/4G)$. The Smarr identity follows from Euler's relation for $\tilde B$, which is homogeneous of degree 2 under $(\Lambda,\sigma,M) \to (\lambda^{-2}\Lambda,\lambda^{-1}\sigma,\lambda M)$. □

The first two terms are the on-shell action of the interior region and the wall. The exterior pseudomanifold contributes only $-\frac{M}{2}\Delta t_E^W$, which tends to zero as $M \to 0$ even though the Euclidean winding diverges. Integrating the first law in M from the $M = 0$ instanton gives $B(M) = S_E^{(0)} + \mathcal{A}(M)/4G - \int_0^M \Delta t_E^W\,dM'$. Numerically the identity holds to 2.5×10^{-8} over 45 models.

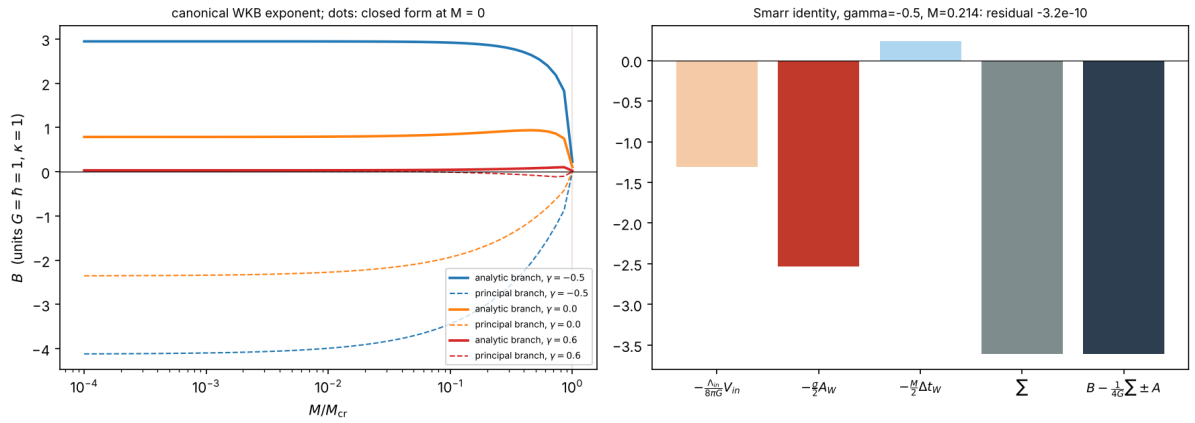


Figure 3: Left: $B(M)$ for the analytic (solid) and principal (dashed) branches; dots: Theorem 5. Right: the terms of the Smarr identity.

8 Numerical results

Over regime A (laboratory data) the ratio $B/B(M \to 0)$ is bounded below by 0.56 for all $\gamma \in [-0.95, 0.95]$, and by 1 for $\gamma \gtrsim 0.2$, where $B(M \to 0) = B_{\mathrm{CdL}}$ (Fig. 3). For GUT-like parameters, $\sigma^{1/3} = \rho^{1/4} = 10^{16}\,\mathrm{GeV}$ ($M_{\mathrm{cr}} \approx 5.6\,\mathrm{g}$), $B = 133$ only for nanogram (sub-Planckian, see Addendum) masses, while $B \approx 7 \times 10^4$ at $10^{-3} M_{\mathrm{cr}}$ and 4×10^6 at $0.5 M_{\mathrm{cr}}$.

9 Discussion and limitations

The three obstructions to universe creation in an asymptotically flat parent, classical (Farhi–Guth), initial-data (bag-of-gold minimal sphere) and semiclassical (Theorem 3), share one origin: the creation of a closed marginally trapped surface in the exterior. Within the reduced canonical theory, the branch of the exponent is fixed and the exponent is covariantly expressible; the remaining ambiguity is located in the Addendum. Open problems: the exact quantum amplitude (an exact solution is possible at $M = 0$, where the constraint becomes an ODE in the momentum representation); whether Lorentzian path integrals select the same contour; prefactors; thick-wall versions of Theorems 3 and 6; non-spherical perturbations. Our novelty claims are provisional pending comparison with the full texts of [3, 4, 7, 8].

Addendum (22 October 2026): corrections and later results

This manuscript reports Stage One only. Later work (Phases II–VI and the roadmap round; the documents named below are in the accompanying repository) corrects or sharpens the following points.

1. *The ambiguity is located, not removed* (PHASE_III). Analyticity fixes the branch of P (Proposition above), but every formulation hides the FGG datum in one place: the boost-angle convention of a flipped normal, equivalently the relative normalisation of the orientation sectors. Exactly one ambiguity survives: in the single-flip class two consistent exponents remain, B_{an} and $B_{\mathrm{sq}} = \frac{2}{G}\int R \arccos|C|\, dR < B_{\mathrm{an}}$, with a universal Stokes multiplier $\sqrt{2}$. The multiplier is now confirmed by exact integration of the momentum-space squared constraint at $M = 0$.

2. *Where the ambiguity lives* (ROADMAP Theorem S4). A laboratory transition is double-flip, hence unambiguous, iff $\kappa < \chi$ and $M < M_{\mathrm{sd}} = (\chi^2 - \kappa^2)/2G\chi^3$; the boundary is

$R_2 = 1/\chi$. In the single-flip class, a sector-independent probability current gives $2e^{-B_{\rm sq}}$, while $e^{-B_{\rm an}}$ requires an entropy-weighted daughter measure (ROADMAP Q9.3–Q9.4).

3. *T5 status* (ROADMAP Part A). For light walls, T5 is a theorem with an explicit universal law $B = B_{\rm CdL}\,\Phi\big((R_{\rm CdL}/R_M)^2\big) \geq 4\pi^2\sigma^2 R_M^2/\rho$. It implies the frontier $B \geq \frac{4}{3}\sqrt{\pi/2}\sqrt{B_{\rm CdL}(M/m_P)(R_M/\ell$ so in any semiclassically controlled regime $P \leq 10^{-23}$. Elsewhere $B \geq 0.564\,B(0)$ remains numerical.

4. *Masses* (PHASE_V). "Picogram" should read nanogram. These masses are below the Planck mass, so the black hole is not semiclassical. For $M \gtrsim 10m_P$ the GUT exponent is $B \gtrsim 7 \times 10^4$.

5. *The daughter needs no boundary condition, and only the Killing clock folds back* (ROADMAP Part D, corrected in Addendum 2). The finite-time "edge" of the daughter channel is a coordinate singularity of the static-frame momentum. Along every daughter $dt_\infty/d\tau < 0$ (region III). The roadmap round read this as "time-reversed daughter, Klein-type channel"; that reading is withdrawn (Addendum 2, item 2).

6. *Literature.* The FMP abstract states that agreement with the Euclidean method requires degenerate metrics [4]; Ansoldi and Tanaka resolve slicing discrepancies with a smooth time slice [17]. Both are consistent with the results above. A full-text comparison is still pending.

Addendum 2 (23 October 2026): the transition amplitude, the daughter's clock, verification

1. *A charge selection rule* (ROADMAP_ROUND3, Theorems A1–A2). Assume (P1) the reduced theory, with the analytic shift constraint $\psi(u + ih) + \psi(u - ih) = 2C\psi$, $u = R^2/2$, $h = \hbar G$; (P2) physical states are geometric and analytic in $|\mathrm{Im}\,u| \leq 2h$, with each configuration represented once; (P3) probability is the conserved Casoratian charge $K[\psi, \psi^\sharp]$, with no entropy weights. Then $K = \Phi(e^{2\pi u/h})$ has a unique Laurent series, and a geometric wave with net flip number n contributes only to q^{-n}. Hence K is constant with ordinary flux balance for double-flip transitions, and $K \equiv 0$ (total reflection) for single-flip ones. For double flip, $P = e^{-B_{\rm an}}(1 + O(\hbar))$. The single-flip daughter receives no flux, and $2e^{-B_{\rm sq}}$ (item 2 above) is superseded under P2. The toy $C = -\lambda u$ is solved exactly by $Z_{-iu/h}(i/\lambda h)$. The Hankel function is the unique solution geometric in both regions, its Casoratian vanishes, and $|H^{(1)}(-u)| = e^{\pi u/h}|H^{(1)}(u)|$, so the decay is the analytic-angle exponent exactly, for every h.

2. *The daughter's clock.* Along every branch, on both sides of the wall, both Kruskal null coordinates increase with proper time, because $|\dot{R}| < |A|$ and $\mathrm{sign}(dt/d\tau) = \mathrm{sign}(A)$. The conserved charge of the expanding double-flip daughter is positive, like that of the incident laboratory wave. Only the exterior Killing time folds back. Reversing its orientation is not a reversal of physical time.

3. *Mechanisms.* Dynamical assistance requires driving frequencies $\omega \gtrsim 1/\Delta\tau_E \approx 0.3$–$0.6\,E_{\rm vac}$. No laboratory mechanism beats the frontier; bubbles nucleated during inflation remain a cosmological route only.

4. *Verification.* State level: invisible (Birkhoff). Event level: on the collapse branch a radiating wall's signal has a late tail $\propto e^{-u/4GM}$ carrying $\Gamma\tau_{\rm collapse}$ photons; on the daughter branch it stops at the turning point, with redshift factor $\sqrt{f(R_1)}$.

5. *Proof gaps narrowed.* T5 strip: $\lim_{E\to E_{AB}^{+}} \Delta t_E^W \propto J_0+\pi<0$ for all γ, from $(z^2+z+1)^2 - z^2h'(z) = (z-1)(z+1)(z^2-z+3)(z^2+z+1)/z^2$. NG4: both boundary values ($F(t)$ and $(|J_0|+\pi)/2\pi$) exceed 1, with the interior monotone numerically. An independent high-precision check gives $B_{\rm an} = B(0)\Phi(E)[1+c(E)\kappa^2/\chi^2]$ with $-0.73<c<0$, so the frontier carries the factor $1-0.73\kappa^2/\chi^2$ ($P \le 10^{-23}$ unchanged). Euclidean Schwarzschild–de Sitter with conical defects at both horizons has $I(\beta) = -(A_b+A_c)/4G$ for every period.

Addendum 3 (23 October 2026): the search for an accessible mechanism

1. *Classical budget.* In spherical symmetry an anti-trapped sphere forms from laboratory data only if, along some ingoing null ray (with $r_v = \frac{1}{2}$ where it leaves the initial cone), $2Gm/r|_0 + 8\pi G\int r(-T_{uu})\,du/\alpha^2 \ge 1$. The NEC must therefore be violated with an integrated null force of order $c^4/8\pi G = 4.8\times10^{42}\,$N, independent of size, or the data must already be near their Schwarzschild radius.

2. *Thin walls of any matter.* At a flip of the exterior orientation $V = f_{\rm out}$, so a classical flip happens inside a horizon. With growing radius its past is a singularity, a Cauchy horizon, or a cosmological region. Exotic matter on the wall can produce only a visible bubble. That needs negative asymptotic tension, destabilises Minkowski space, and consumes the parent.

3. *Routes without exotic matter.* A charged exterior changes the exponent by $< 3\%$ but yields a classical route through the Cauchy horizon above $M_{\rm cr} \approx 0.16\,m_P^3/\sqrt{\rho}$. A stabilised seed would convert in e^B/ω. A tilted-quartic field theory shows that the frontier $B_{\rm CdL} = 10$ needs $\lambda \sim 10^4$. With $\lambda \le 4\pi$, $P \le 10^{-476}$.

4. *The only laboratory-scale class* is self-amplifying stable NEC violation (Galilean genesis and its relatives). It meets the budget automatically and forms a daughter deterministically. It requires violation of the GSL (Wall's theorem) and of S-matrix positivity. A seed bounded by a thin interface in vacuum becomes a visible, parent-consuming bubble, because the interface energy turns negative at $H^2R^2 = 2GM/R$. A hidden daughter needs a throat to open inside a NEC-violating boundary layer.

5. *Observables.* For an explicit 1+1 field in the laboratory state, the collapse branch shows a negative precursor and a switch-on of Hawking emission over $\approx 3.5\times 4GM$. Under a stated state assumption the daughter branch differs by ≈ 0.1 quanta per event. The daughter's inflation is not observable from outside.

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